SAMUEL BONNEY

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Research Interests

Nonparametric inference, goodness-of-fit testing, kernel density estimation, time series analysis, and biostatistics.

Education

- **Master of Science, Applied Statistics** *Aug 2025 – May 2027 (expected)*
  Illinois State University Normal, IL
  GPA: 3.87/4.00
  Relevant coursework: Applied Probability Models, Applied Stochastic Processes, Statistical Computing, Advanced Linear Algebra, Regression & Time Series Analysis (in progress)

- **Bachelor of Arts, Mathematics and Statistics** *Sep 2019 – Sep 2023*
  University of Ghana Accra, Ghana
  GPA: 3.64/4.00 · First Class Honours
  Relevant coursework: Integration Theory and Measure, Statistical Inference I & II, Multivariate Distributions, Nonparametric Statistics, Regression Analysis, Complex Analysis

Honors & Awards

- **Outstanding Master's Student,** Illinois State University, 2026
- **Kenneth Berk Scholarship,** Illinois State University, 2026
- **JSM Diversity Mentoring Program Award,** American Statistical Association, 2026
- **Fostering Diversity in Biostatistics Workshop Award,** ENAR, 2025 (for the March 2026 workshop)
- **Winner, Statistics and Actuarial Science Quiz Competition,** University of Ghana, 2023
- **Winner, Mathematics Society Quiz Competition,** University of Ghana, 2023

Research Experience

- **Graduate Thesis** *Aug 2026 – Present*
  Nonparametric Tilted Density Estimation and Bickel–Rosenblatt Goodness-of-Fit Testing
  Advisor: Prof. Fuxia Cheng, Illinois State University
  ◦ Investigating tilted kernel density estimators with cross-validation-based bandwidth and probability-weight selection, using simulation studies to evaluate asymptotic behavior, empirical test size and power, and performance relative to conventional kernel density estimators and Bickel–Rosenblatt goodness-of-fit procedures.

- **Independent Research Project** *May 2026*
  County-Level Lung Cancer Mortality in the United States
  ◦ Integrated CDC SVI, CDC PLACES, and NCI data for 2,536 U.S. counties and applied robust OLS, quantile regression, Moran's I, LISA, spatial error modeling, random forest, XGBoost/SHAP, and mediation analysis, identifying 232 unexplained high-mortality spatial clusters and estimating that COPD mediated 50.8% of the county-level smoking–mortality association.

- **Independent Research Project** *March 2026*
  County-Level Food Insecurity and Chronic Disease Burden in the United States
  ◦ Analyzed 2,299 U.S. counties using food insecurity, social-needs, and chronic-disease data; constructed a standardized disease-burden index and applied multivariable regression, sensitivity analysis, spatial autocorrelation, and geographic mapping to identify 408 counties with jointly high food insecurity and chronic disease burden.

- **Research Project** *Jan 2024 – Aug 2024*
  Modeling the Prevalence of Malaria Among Pregnant Women in Ghana Using Explainable Machine Learning Techniques
  ◦ Applied machine learning algorithms to Demographic and Health Survey data to predict malaria risk; the k-nearest neighbor model achieved the highest performance (AUC = 0.88), and feature-importance analysis ranked factors such as age and anaemia status.

- **Undergraduate Thesis** *Jan 2023 – Sep 2023*

  Application of Machine Learning Models to Identify the Risk Factors of Incidence of Malaria Among Children Under Five Years of Age in Ghana

  Advisor: Dr. Gabriel Kallah-Dagadu, University of Ghana

  ◦ Evaluated five machine learning algorithms, including random forest and neural networks, to predict malaria incidence using the 2019 Ghana Malaria Indicator Survey, identifying key risk determinants for children under five.

Selected Projects

- **BioMed-Scribe: Medical Specialty Classification from Clinical Conversations** *Jan 2026 – Apr 2026*
  ◦ Fine-tuned GPT-2 Medium on 13,492 clinical conversations using two-block parameter-efficient fine-tuning (25.2M trainable parameters; 7.1% of 354.8M total), achieving 79.1% test accuracy and 0.799 macro-F1 across cardiology, neurology, orthopedics, and general medicine, with t-SNE embedding analysis and deployment through an interactive Gradio application.

- **Statistical Analysis of Diabetes Risk Factors Among American Adults** *Jan 2026 – May 2026*
  ◦ Analyzed 253,680 CDC BRFSS observations using SAS and R, applying chi-square tests, factorial ANOVA, multiple and logistic regression, model selection, diagnostics, and sex-stratified analyses to identify clinical, behavioral, demographic, and socioeconomic factors associated with diabetes-related outcomes.

- **Gender-Based Analysis of Cardiovascular Disease Risk Factors** *Dec 2025*
  ◦ Analyzed 68,672 adults in Minitab using nonparametric tests, chi-square tests, and binary logistic regression with gender-specific models; the combined model achieved an AUC of approximately 0.79, with age, BMI, blood pressure, and cholesterol as key predictors.

- **Modeling Life Expectancy in Africa Using Socioeconomic and Health Indicators** *Oct 2025*
  ◦ Built a multiple linear regression model of life expectancy from socioeconomic and health indicators, with variable selection and diagnostics, identifying adult mortality and HIV incidence as the strongest predictors.

Teaching & Work Experience

- **Graduate Teaching Assistant**
  Department of Mathematics, Illinois State University Aug 2025 – Present
  ◦ Lead weekly discussion and problem-solving sessions for MAT 113 (Elements of Mathematical Reasoning), MAT 119 (College Algebra), and MAT 120 (Finite Mathematics).
  ◦ Grade assignments and examinations and give written feedback to students.

- **Teaching and Research Assistant (Statistical Consulting Unit)**
  Department of Statistics and Actuarial Science, University of Ghana Sep 2023 – Aug 2024
  ◦ Led tutorial and review sessions for undergraduate and graduate students, including STAT 223 (Elementary Statistical Methods) and STAT 456 (Actuarial Statistics).
  ◦ Supported faculty research in the consulting unit through data cleaning, literature reviews, and exploratory data analysis for study design.

- **Enumerator and Data Analyst**
  Ghana Baseline Evaluation Survey Jan 2024 – Feb 2024
  ◦ Collected field data across multiple sampling units, following quality-assurance protocols to minimize entry errors.
  ◦ Analyzed survey data to support monitoring and evaluation of program effectiveness.

- **Data Analyst Intern**
  Perception Management International Limited Sep 2022 – Dec 2022
  ◦ Built pricing models using SQL and Excel for 50+ due diligence engagements across 9 sectors, including finance, energy, and telecommunications.
  ◦ Designed automated Power BI dashboards that cut weekly manual reporting time by 50% (from 8 to 4 hours).
  ◦ Contributed data-driven market research to 6 successful client proposals.

Presentations & Conferences

Joint Statistical Meetings (JSM) and Diversity Mentoring Program, American Statistical Association, Boston, MA *Aug 2026*

County-Level Lung Cancer Mortality in the United States: Social Vulnerability, Behavioral Risk Factors, Spatial Clustering, and Unexplained Mortality Hotspots, Research Presentation, Illinois State University *May 2026*

County-Level Food Insecurity and Chronic Disease Burden in the United States: Identifying Priority Counties for Intervention, Research Presentation, Illinois State University *March 2026*

ENAR Fostering Diversity in Biostatistics Workshop, Indianapolis, IN *March 2026*

Application of Machine Learning Models to Identify the Risk Factors of Incidence of Malaria Among Children Under Five Years of Age in Ghana, Departmental Seminar, University of Ghana *Nov 2023*

Skills

- **Statistical Software:** R, SAS, Stata, SPSS, Minitab
- **Programming & Databases:** Python, SQL
- **Data & Visualization:** Power BI, Excel, Geographic Mapping
- **Documentation & Version Control:** LaTeX, R Markdown, Quarto, Git, GitHub

Professional Memberships

Institute of Mathematical Statistics *Sept 2025 – Present*

American Mathematical Society *Oct 2025 – Present*

Leadership & Service

MathEye Impact, Co-Founder / Vice President *2021 – Present*

- Co-founded and help lead a non-profit boarding school for high-achieving students from low-income families across Ghana, mainly the Central Region; the school now serves 275 students.
- Inaugural class achieved the best results in Ghana on the Cambridge International Examinations, with A/A* grades on over 70% of exams taken and 5 students ranked first nationally in their subjects.

Treasurer, Ghanaian Students Association, Illinois State University *2026 – Present*

- Manage the association's budget, financial records, reimbursements, and expenditure tracking, and work with the executive team on event planning.

Solution Space Member, The Mathematics Society, University of Ghana *2020 – 2023*

- Helped fellow students work through challenging mathematics problems and build independent problem-solving strategies.